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MAGNETIC POLARITY STRATIGRAPHY OF THE MIOCENE RIO AZUL SECTION, PRECORDILLERA THRUST BELT, SAN JUAN PROVINCE, ARGENTINA[1]

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ABSTRACT

The chronostratigraphy of synorogenic strata of Río Azul in the Bermejo foreland basin, Argentine Andes, reveals details of Miocene tectonic activity and deposition. Chronostratigraphic interpretations are based on magnetic polarity stratigraphy and zircon fission-track dates of interbedded volcanic ashes. The Río Azul section is located within the Precordillera thrust belt, uplifted and rotated by motion on subjacent faults. The Cenozoic strata are themselves broken by two principal zones of faulting, whose displacements are great enough to affect correlation of the local magnetic stratigraphy to the magnetic polarity time scale. The 3200 m thick section coarsens upward, from sandstone and mudstone (Cuculí Formation) to conglomerate (El Corral Formation); the magnetostratigraphic approach could be used only in the relatively fine-grained strata of the lower part of the section. The radiometric ages indicate the entire section is of middle to late Miocene age, and we correlate the magnetic polarity pattern of each fault block to the magnetic polarity time scale. The lower 500 m of the section apparently ranges from approximately 16 to 12 Ma, although the age of the lowest 100 m is quite uncertain. The fault-bounded block from 500 to 1000 m probably correlates to approximately 13.5–12.5 Ma. The block from 1000 to 2000 m correlates to approximately 12.5–10 Ma. Given the lack of unconformities in the coarse uppermost 1200 m of section and a zircon fission track age within the conglomerates, we speculate that the upper part of the section was deposited very rapidly; the top of the section is probably no younger than about 8 to 9 Ma. These correlations and similarities of facies, paleocurrents, and provenance imply that the Cuculí Formation represents the proximal facies correlative to the Jarillal Formation of Sierra de Huaco, and that the El Corral conglomerates are the up-dip equivalent of the Huachipampa Formation of Huaco. These results are consistent with a paleogeographic model of a major drainage reorganization in the Bermejo foreland basin at about 10 Ma. It is clear, however, that thrusting in the Precordillera began more than 2 m.y. prior to the drainage reorganization.

INTRODUCTION

Foreland basins contain the most complete record available of the deformation and uplift history of mountain belts, although such histories must be extracted from among the other signals recorded in sedimentary rocks. The Miocene to Recent Bermejo foreland basin, San Juan Province, Argentina, formed as the Precordillera thrust belt of the Andes Mountains deformed (Jordan et al. 1989). We have focused study on the Bermejo basin in an attempt to document and understand the relationships among shortening, magmatism, and subduction in the region of flat subduction of the Nazca plate beneath the South American plate (Jordan et al. 1983).

The Río Azul section is part of a structural sliver of the western portion of the Bermejo basin. Río Azul is one of the few thick (3200 m), laterally and vertically continuous Bermejo basin remnants exposed within the central Precordillera thrust belt (fig. 1). This study of the older, more proximal margin of the basin complements work in the extensive exposures of more distal Bermejo basin strata farther east, in Sierra de Huaco and Sierra de Mogna (Johnson et al. 1986; Beer 1990; Milana, Bercowski and Jordan unpub. data).

The oldest Neogene strata known in the Bermejo basin accumulated approximately 17 to 19 Ma (Reynolds et al. 1987; Tabbutt et al. 1987). Alluvial facies dominate all Bermejo basin sections, with principal source areas to the west in the Frontal Cordillera and Precordillera (Damanti 1989). Sections exposed within the central Precordillera are older than

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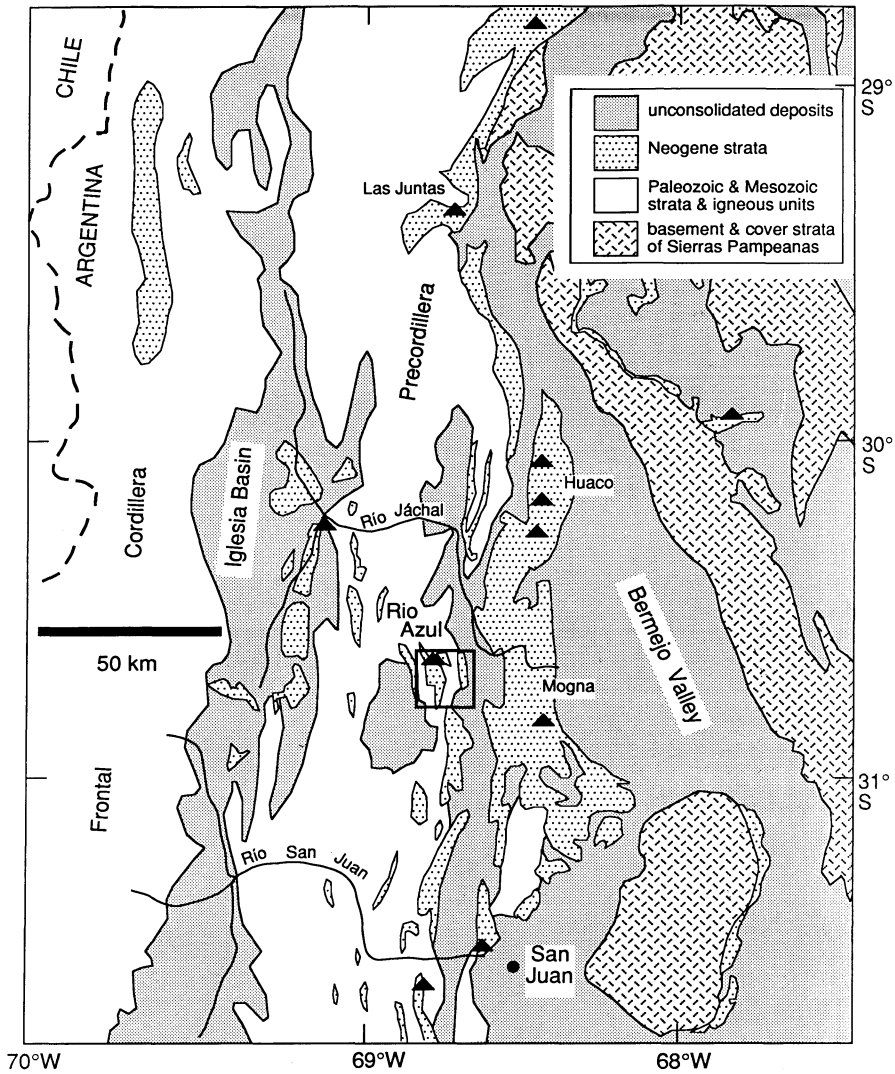


FIG. 1—Location of Río Azul study area (box) in Precordillera thrust belt, and neighboring major geologic provinces. Triangles indicate sites where magnetic polarity studies have been completed or are in progress (Bercowski et al. 1986; Johnson et al. 1986; Johnson et al. 1987; Beer 1990; Reynolds 1987; Reynolds et al. 1987, 1990; G. Re and J. Vilas unpub.; J. Reynolds, G. Figueroa and F. Bercowski unpub.; J. Milana, F. Bercowski and T. Jordan unpub. data).

those in the eastern Precordillera, implying a west-to-east progression of accumulation and subsequent uplift. Exposures in Sierra de Huaco indicate that a major change in the drainage system feeding the fluvial system occurred ~10 Ma (Damanti et al. 1988; Beer and Jordan 1989). Our purposes for study of the Río Azul section were to improve the paleogeographic model of the drainage evolution of the Bermejo basin, to define better the three-dimensional geometry of the basin fill, and to obtain more direct information about the times of deformation of major fault sets in the Precordillera.

Magnetic polarity stratigraphy coupled with zircon fission-track tephrachronology have proven efficient and powerful tools of chronostratigraphy in Neogene strata of west-central Argentina (Johnsson et al. 1984; Johnson et al. 1986; Reynolds, 1987). The Río Azul study further demonstrates the suitability of the strata to these techniques; 29

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polarity zones were identified and three volcanic ashes were dated. Nevertheless, study of a section exposed by virtue of thrust faulting in the midst of a thrust belt tempts fate. An essential premise of correlation of a local magnetic polarity stratigraphy to a magnetic polarity time scale is that the magnetostratigraphic record is complete and not repeated. At Río Azul, the section is broken into three fault blocks, thus violating that premise. However, because ash ages are available in two of the fault blocks and because there are several polarity zones per fault block, correlations can be made for each fault block independent on the other fault blocks.

This study also illustrates the challenge of utilizing magnetic polarity stratigraphy for chronology of middle Miocene units. The high frequency of polarity reversals between 10 and 15 Ma (~1 per 208,000 yr) (Berggren et al. 1985) offers the appealing possibility of unusually precise age control. However, to achieve a correct correlation between a local polarity column and the magnetic polarity time scale is more difficult than in intervals of the Cenozoic with less frequent reversals—independent biostratigraphic or radiometric data with accuracies and precisions approaching ± 100,000 yr must be employed to tie the local reversal pattern to the magnetic polarity time scale. In the absence of such independent chronologic control, we must assume that the rate of accumulation in the local column was either constant through time or varied in a simple fashion, and therefore that similarities of pattern between the local polarity pattern and the magnetic polarity time scale are a meaningful basis for correlation.

Because of the faults and the high frequency of middle Miocene polarity reversals, determination of the ages of the Río Azul strata proved to be more challenging than in previous studies of the Bermejo basin. This is an example of good quality magnetostratigraphic data but geologic complexities; therefore we describe in depth the paleomagnetic methods and results, and the premises of our correlations.

GEOLOGIC AND STRATIGRAPHIC SETTING

The Precordillera thrust belt is characterized by Paleozoic strata that trend nearly north-south and are imbricated on west-dipping thrust faults (Baldis and Chebli 1969; Allmendinger et al. 1990). The age of deformation in the Precordillera is constrained by cross-cutting relations with Neogene strata and by interpretation of synorogenic strata. The deformation age is broadly middle Miocene to Recent, and available information suggests that the age of the eastern thrust margin migrated from west to east (Johnson et al. 1986; Beer and Jordan 1989; Reynolds et al. 1990). Nevertheless, the exact time of movement(s) of most of the thrust sheets is not yet known. Determination of the ages of Neogene strata preserved in rare structural slivers within the thrust system holds considerable promise for improving our knowledge of the deformation chronology.

The study area is located in the eastern part of the severely shortened central portion of the Precordillera (Allmendinger et al. 1990). In this structural sliver the Neogene units are exposed continuously for about 20 km along strike and are cut by the Río Azul. The Neogene units and underlying Mesozoic(?), Permian and Carboniferous units are rotated westward (about a horizontal axis) in the hanging wall of a thrust that places Carboniferous units over another structural sliver of Neogene strata (fig. 2, fault A). The western margin and top of the stratigraphic section is also bounded by a thrust fault, which places units as old as the Ordovician over Neogene strata (fig. 2, fault B). Where examined, the fault (B) bounding the top of the section dips 30° more steeply than bedding in the footwall, whereas the fault (A) on the east side of the structural sliver is approximately parallel to steeply dipping strata in its footwall. In addition, other thrust faults occur with considerable stratigraphic repetition less than 10 km to the east and west of the studied section (fig. 2, faults C and D), and Neogene strata 10 km east of the measured section are probably exposed by deformation above a blind thrust (fig. 2). Thus a minimum of five thrust plates moved within the neighborhood of the Río Azul section, but their subsurface geometries and possible interrelationships are not known. Based on cross-cutting relations, final movement on the fault on the western edge of the measured section (fault B, fig. 2) must post-date the youngest units in the section, and all the deformation that rotated the measured section and pre-

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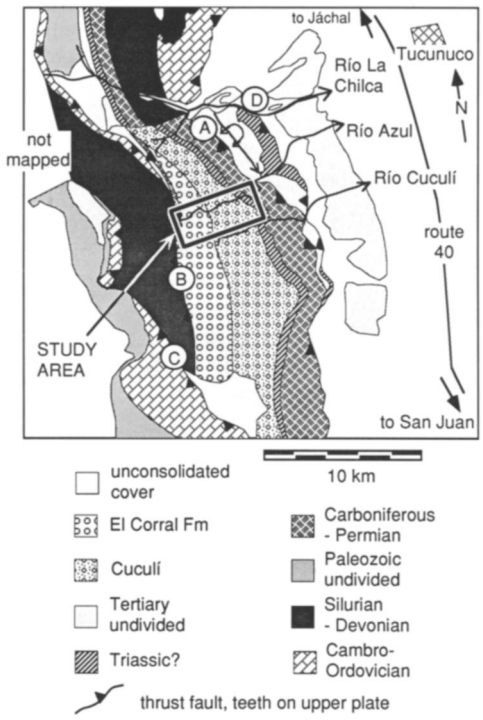


FIG. 2.—Geologic map of Río Azul structural sliver, in the central Precordillera thrust belt (see box, fig. 1). Letters indicate specific thrust faults referred to in text.

sumably rotated the thrust at the east side of this structural sliver (fault A) must post-date all of the Río Azul strata.

Neogene strata elsewhere in the Bermejo basin are entirely nonmarine in origin, representing a variety of fluvial, alluvial fan, and lacustrine environments in an arid setting. The base of the Bermejo basin section is dated precisely only at Las Juntas, 150 km north of Río Azul, where it is ~18 Ma and lies above Triassic (?) strata on an erosional unconformity (Reynolds et al. 1987, 1990). At other locations the base is approximately 15 to 19 Ma (Tabbutt et al. 1987, 1989; Reynolds 1987; Reynolds et al. 1988). The uppermost strata of the Bermejo basin that have been deformed and dated are as young as about 2 Ma (Johnson et al. 1986; Jordan et al. 1988; Milana, Bercowski and Jordan unpub. data). Although strata accumulated in the Bermejo basin during the Quaternary, they are not yet exposed and dated.

Furque (1983) divided the Tertiary strata of the Río Azul structural sliver into two formations, the Río Cuculí and El Corral (figs. 3, 4). According to Furque (1983), the Río Cuculí Formation consists of a mixture of sandstone, siltstone, and conglomerate of fluvial channel and overbank origin. The overlying El Corral is dominated by conglomerate and constitutes proximal bajada deposits. Furque (1983) designated the Río Francia, about 6 km north of Río Azul (fig. 2, south fork of the river system labeled Río La Chilca), as the stratotype of the Cuculí Formation, where he subdivided it into three members. However, along Río Azul those members are not distinctive lithologic packages. It is interesting to note that Furque (1983) assigned a different stratigraphic terminology to Tertiary units in the footwall of fault A (fig. 2), which implies they are more typical of Neogene units exposed farther to the east in the vicinities of Huaco and Mogna.

The base of the Cuculí Formation is unconformable above the Triassic (?) Vallecito Formation. The Vallecito consists of eolian sandstones and can be traced readily at least 150 km to the north. The Vallecito could be as old as Triassic or as young as Cenozoic (Furque 1983). The contact between the Cuculí and El Corral Formations is conformable and gradational along the Río Azul, although Furque (1983) describes a strong erosional unconformity between similar lithologies elsewhere in the area. The El Corral Formation is overlain by Carboniferous-Permian redbeds and those in turn by the Devonian Punta Negra Formation across steeply inclined thrust faults.

METHODS

Our tasks were reconnaissance mapping, section measurement, magnetic polarity stratigraphy, tephrachronology, facies analysis, paleocurrent analysis, and petrography. Base materials utilized in reconnaissance mapping included Thematic Mapper images and Large Format Camera photographs acquired from the space shuttle, on which the major formations could be easily distinguished and traced, low-altitude stereoscopic airphotos, and Furque's (1983) geologic map (1:200,000). We measured the Río Azul section with Brunton and tape.

Oriented samples for paleomagnetic measurement were collected initially at sites averaging 22 m stratigraphic separation in the lower 2100 m of the section. Following the

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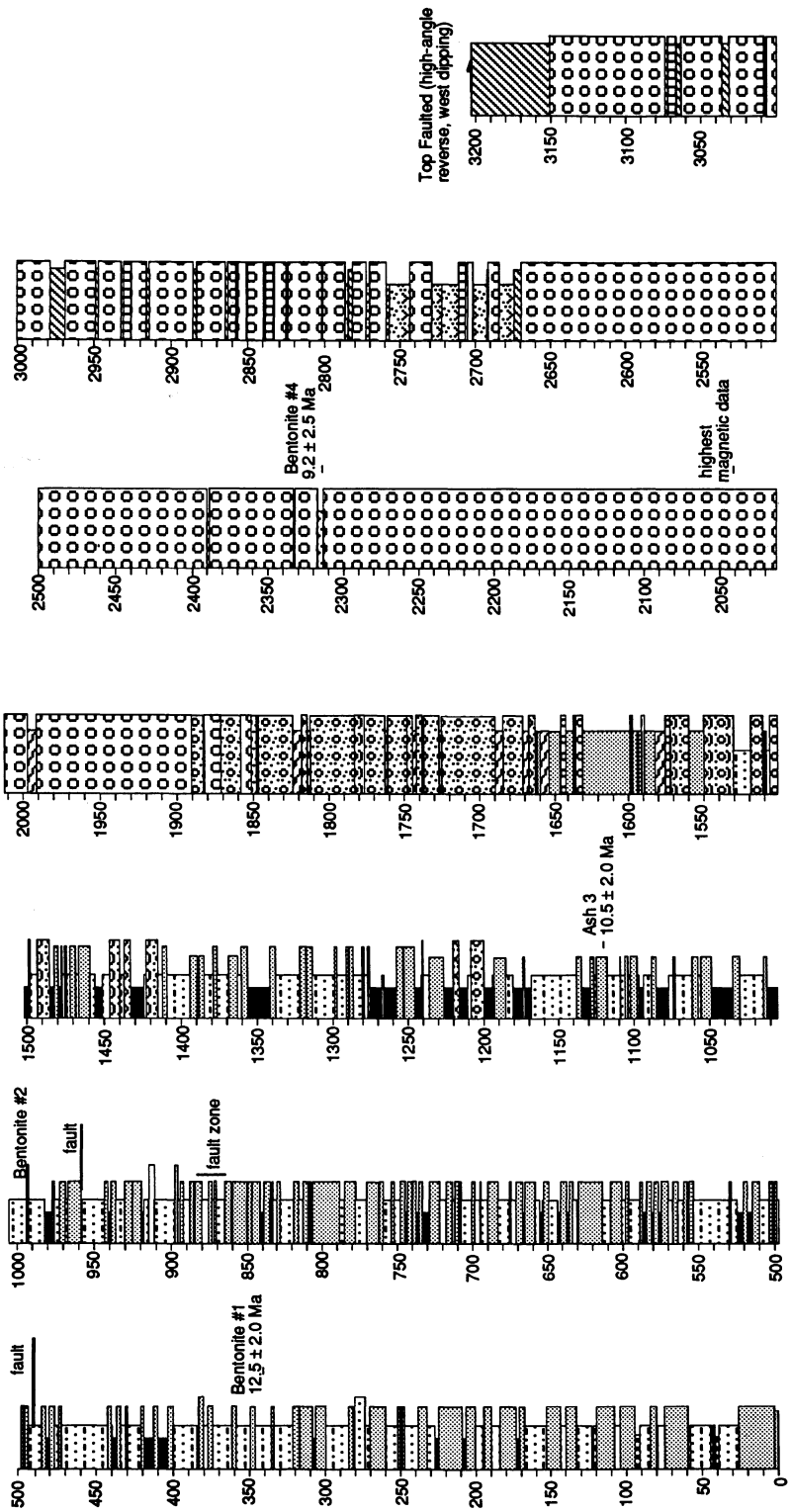


FIG. 3.—Lithologies of the Río Azul measured section, which is the basis for the cartoon column of figure 4*a*. Width of pattern increases with grain size of sediment. Levels of four primary volcanic deposits shown with the ages of the three that yielded zircon separates.

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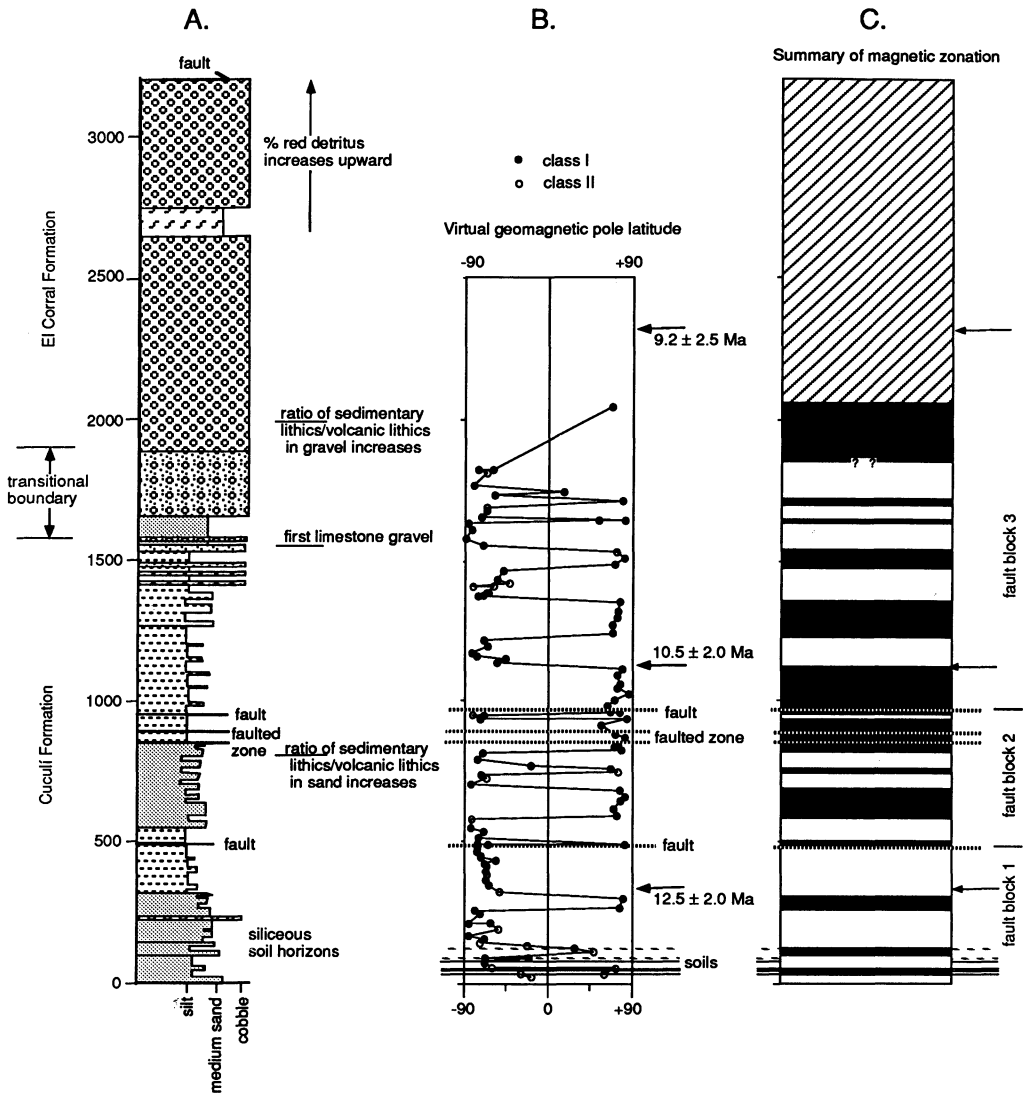


FIG. 4.—(*a*) Simplified stratigraphic section for Río Azul, showing formations, basic textures, faults, and key petrographic horizons (see text). (*b*) Stratigraphic levels and VGP latitudes of class I and class II paleomagnetic data. The locations of dated volcanic beds, and their ages, are indicated with arrows. (*c*) Composite magnetic polarity zonation of the Río Azul section, derived from the VGP latitude plot (b). Black zones have normal polarity, and white zones have reversed polarity. Note subdivision of magnetic section into three fault blocks. The upper 1200 m of section (diagonal ruled) lack data for magnetic polarity zonation.

method established by Johnson et al. (1975), three fist-sized samples of suitable fine-grained lithologies were collected at each site, wrapped carefully, and taken to the laboratory. The oriented samples were ground into 2.5 cm cubes on a disk grinder and the paleomagnetism was then measured with a Schonstedt SSM-1 spinner magnetometer in the Dartmouth College paleomagnetism laboratory. Based on the results of magnetic polarity determinations from the first year, we returned to the field to sample intervals for which the original samples were destroyed or for which the preliminary magnetic zonation was ambiguous. The final magnetic zonation, presented here, is based upon 112 sites in the

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lower 2040 m of section, with an average site spacing of 18 m (fig. 4*b*). Given the very coarse-grained nature of the upper 1200 m of the section and nearly complete lack of fine-grained interbeds, it was impossible to collect paleomagnetic samples in the upper part of the section.

Magnetic polarity stratigraphy has been utilized in several other studies to date the unfossiliferous nonmarine strata of the Bermejo and other basins in western Argentina. In many cases the magnetic carrier of remanence appears to be detrital magnetite, and determination of detrital remanent magnetization (DRM) was relatively straightforward (e.g., Johnsson et al. 1984; Johnson et al. 1987). In other sequences hematite is the common remanence carrier, and isolation of the primary magnetization requires extensive demagnetization (Reynolds 1987; Reynolds et al. 1990; Johnson et al. 1986; Beer 1990). The standard procedure followed in those other studies and utilized here is to determine the average paleomagnetic vector orientation of a minimum of three samples from each site, using standard statistics (Fisher 1953). This is accomplished through stepwise demagnetization of a minimum number of samples (six in this study) to establish a simplified laboratory treatment ("blanket treatment") as a *starting point* for all of the other samples. For all samples the natural remanent magnetization (NRM) is first measured. Each sample is then carried through progressive demagnetization steps and remeasured. This procedure usually succeeds in unambiguous determination of the direction of the characteristic remanent magnetism and establishes that no additional magnetic components exist.

In Neogene sections of Argentina, Bolivia, and Pakistan which have simple magnetization properties, results based on the simplified laboratory treatment just described have been replicated by more thorough studies. In some cases duplicate sections have been studied short distances along strike (Johnson et al. 1988; McRae 1989, 1990; Beer 1990); in others stepwise thermal demagnetization was carried out for every site (McRae 1990), and in some cases additional sites were collected to increase the stratigraphic density of sampling (Johnson et al. 1988; McRae 1990). In all these cases, the additional labor has reconfirmed the validity of the initial, simplified approach: the major zonation pattern of the magnetic column has been reproduced, although details of the stratigraphic placement and character of the magnetic transitions have been refined.

Four several-kilogram samples of pure volcanic ash or bentonite were collected within our measured section (fig. 3); three produced datable zircons. Ash samples were ground to <60 mesh using a jaw-crusher followed by a disc-pulverizer, whereas bentonite samples were hydrated and sieved. To produce clean zircon separates, a Wilfley table, Frantz magnetic separator, and heavy liquids were employed. The zircons were dated using the external detector method (Fleischer et al. 1964; Naeser and Dodge 1969; Naeser 1978), and the ages and errors were calculated using the methods described by Naeser (1978), Johnson et al. (1979), and McGee et al. (1985). The thermal neutron flux was determined by irradiating NBS glass SRM962 with the samples. Dr. Charles Naeser calibrated the neutron standards.

THE RÍO AZUL SECTION

Three fault zones are recognized in the Río Azul section whose offsets are large enough that units cannot be matched across the fault at the scale of single outcrops; these zones occur at 490, 865, and 960 m above the base of the Neogene section. The faults, or commonly pairs of faults, are associated with folds whose axes typically trend northward, parallel to fault trace. The faults trend N 30 to 40 W and dip westward, but 20°–30° less steeply than the bedding in the adjacent strata. The fault geometries are not like those of the thrust faults that bound the Río Azul structural sliver. Although the fault zones are large enough that they could be traced in the field southward to the Río Cuculí (fig. 2), we do not know from field relations or from lithologic sequence whether they repeat or omit section and whether the motion was dip-oriented or oblique. In the absence of this information, we must depend upon the magnetic polarity zonation to constrain the fault motions. For convenience we will speak in terms of three fault blocks: the lower fault block (1) has structurally continuous strata beneath 490 m, the upper fault block (3) has structurally continuous strata above 960 m, and the middle fault block (2) has strata be-

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tween 490 and 960 m that are further broken by at least one other fault.

Determining the base of the Neogene section is problematic. The Vallecito Formation consists of purplish-brown medium to coarse sandstone with large-scale cross beds of likely eolian origin (Furque 1983), and the lower Cuculí Formation has interbedded tan channel sandstones and deep red pedogenically altered overbank siltstone. Between these two clearly differentiable lithofacies is a 10 m thick interval of red medium-grained sandstone with large trough-cross beds. We placed the Vallecito-Cuculí contact at the top of the purplish eolian beds.

The lower 1700 m of the Río Azul section constitute the Cuculí Formation (Furque 1983) and the overlying El Corral Formation is 1500 m thick. The Cuculí Formation is dominated by tan to reddish brown sandstones and siltstones (fig. 3). In the basal few hundred meters, thick sandstone beds are common (fig. 5*a*) with local bar morphologies, lateral "wings" extending out from the central channel region, and lateral accretion surfaces that pinch into mud-filled hollows. Mudcracks are common and cross beds are present but not common. These facies are interpreted to represent deposits of modest relief, moderately sinuous, fluvial channels; some of the sandstones constitute multistoried channels. Higher in the Cuculí Formation the sandstones tend to be more thinly bedded and laterally continuous over distances of ~100 m, with local erosional relief of 1 m at the bases of beds. These are deposits of broad, shallow fluvial channels. Another lithofacies of the Cuculí Formation consists of thickly bedded, horizontally laminated sandstones. These are interpreted to be sheetflood deposits. Thick horizons of thinly interbedded fine siltstone to claystone and fine sandstone are interpreted to be overbank deposits. The scale of bedding and channel forms is greatest in the lower few hundred meters of the section (fig. 5*a*). To date, few paleocurrent data have been collected, but east-west paleoflow is suggested by ridges on the bases of sandstone beds, north-northeast-directed paleoflow is indicated by flute casts, and southeast-northwest flow is implied by primary current lineation.

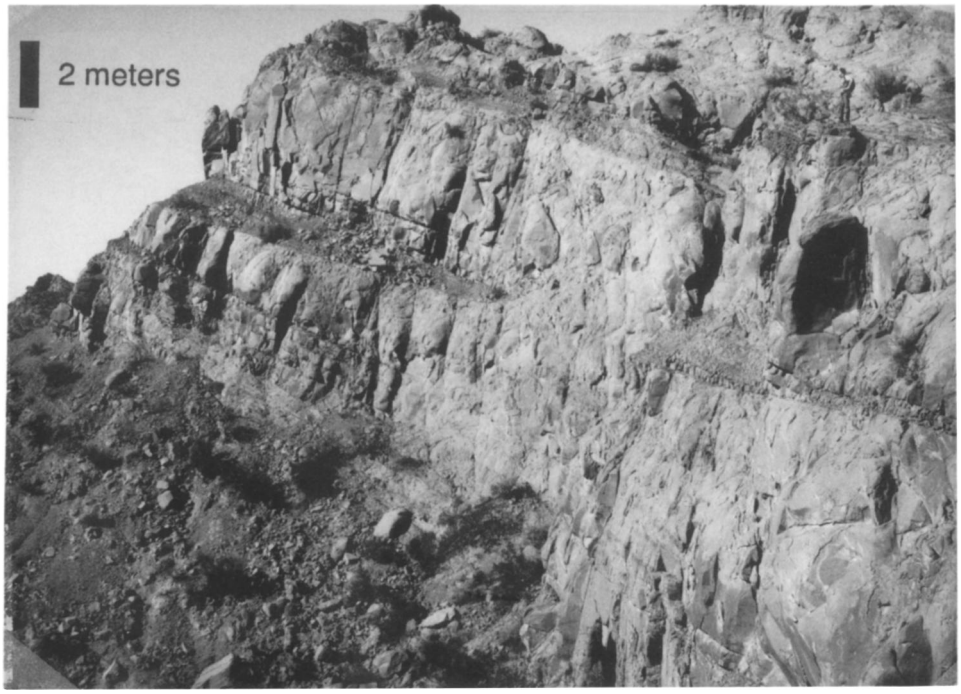
There is little evidence of pedogenic alteration of the finer-grained deposits of the Cuculí Formation, and commonly primary laminations are preserved in spite of partial mixing by burrowing. However, there are noteworthy exceptions to this generalization in the lower 130 m of the section (figs. 3, 4), where six distinctive soil horizons occur. The lower four horizons, each roughly 1 to 5 m thick, are characterized by bright red to dark reddish-brown siliceous zones with gray mottling, bioturbation, and local blocky texture. The upper two zones are orange colored and less indurated, suggesting less mature soil formation. Mudstones above 400 m (fig. 4) include some interbeds with incipient soil development.

The lowest conglomerate bed in the measured section occurs at about 220 m and is composed largely of pebbles derived from siliceous volcanic rocks. Pebbly bases of sandstone beds have similar compositions, plus green low-grade metasedimentary clasts. The upward-coarsening trend of the upper Cuculí Formation begins at about 1200 m, where a conglomerate with maximum clast size of 15 cm occurs. Throughout the remainder of the Cuculí Formation the maximum clast size of conglomerate beds is typically 10 to 15 cm. The conglomerate beds are internally organized into 5 to 20 cm thick laminae of varying grain size; the beds define channels with as much as 2 m erosional relief at their bases and with thin lateral "wings."

The Cuculí lithofacies are not identical to those of Neogene sections in regions east and north of the zone with which we are familiar (e.g., Reynolds et al. 1990; Johnson et al. 1986; Beer and Jordan 1989). There are, nevertheless, many characteristics in common between the strata in the lower 1200 m and the Quebrada del Jarillal Formation of Sierra de Huaco exposed to the northeast of Río Azul (fig. 1). Channels and bar forms of the Río Azul area are, however, larger-scale than those of Huaco. In general, we interpret strata of the lower part of the Cuculí Formation to have been deposited on a channelized alluvial plain with only modest depositional relief and ephemeral discharge. Floods sometimes produced channelized deposits characterized by lateral accretion surfaces, and other times resulted in sheetflow deposits.

The transition from Cuculí to El Corral facies occurs in a zone about 300 m thick.

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A



B

FIG. 5.—(*a*) Photograph, looking south, of sandstone beds of lower part of Cuculí Formation, illustrating tabular nature and amalgamation. (*b*) Photograph, looking south, of conglomerates of El Corral Formation that are included ~50° to the west (inclined black bar in foreground parallels bedding in outcrop). Foreground shows gravel of modern stream bed. The 5 cm to 1 m thick, well-defined but not sharply bounded stratification is characteristic.

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The color of the conglomerates, their fabric, and the nature of fine-grained interbeds all are transitional. Channelized conglomerates typical of the Cuculí Formation are interbedded with sheet-like conglomerate bodies in the transition zone. These sheet conglomerate beds include up to 50% coarse sandstone. Thickly interbedded with the conglomerate beds are fine- to medium-grained, laterally continuous, orange sandstones.

The El Corral Formation is dominated by clast-supported conglomerates with well-developed sheet-like stratification (fig. 5*b*). Only about 1% of the beds are sandstone. Whereas the general aspect of conglomerates in the Cuculí Formation is light colored and the clasts are of many hues, the El Corral is characterized by its medium-gray overall tone and largely green clasts. Typical El Corral clasts are coarse pebbles to cobbles, with local boulders. Horizontal, planar lamination is ubiquitous; clast-size variations define laminae 5 cm to 1 m thick (fig. 5*b*). Normal grading of clast size occurs locally. Clast fabric is moderately organized, with common imbricated zones. At some horizons calcite cement seems to displace clasts, suggestive of soil-zone calcrete. Local truncation defines only minor erosional relief between conglomerate beds (15 to 100 cm). Clast size increases progressively upward in the El Corral Formation. At the base of the El Corral, the maximum clasts are 30 to 40 cm diameter, 500 m higher in the section the average maximum clast is 45 to 50 cm in diameter, and in the uppermost 200 m the average maximum clast exceeds 90 cm diameter.

Red sandstones and conglomerates first appear as discrete thin interbeds at 2390 m in the section (fig. 3). These redbeds contain abundant clasts of red sandstone, indicating that their coloration is of detrital origin, rather than from diagenetic alteration. The redbeds increase in abundance upward until they dominate the upper 200 m of the El Corral Formation. The detrital redbeds also coarsen upward from mostly sandstone to coarse conglomerate. These conglomerates do not have the well-ordered fabric typical of the intercalated gray conglomerates and are less continuous laterally.

Imbricated clasts define paleocurrents in the El Corral Formation. To avoid potential bias caused by the orientation of outcrop faces, at six arbitrarily selected levels in the El Corral Formation the orientation of a–b planes of 10 blade-shaped clasts was measured. This was done without any regard as to whether or not the clasts appeared to be imbricated, but the rare cross-bedded horizons were avoided. Those clasts define a mean paleoflow vector oriented N97E; the three sites with best clustering of data, implying true imbrication, define a paleoflow orientation of N82E (fig. 6).

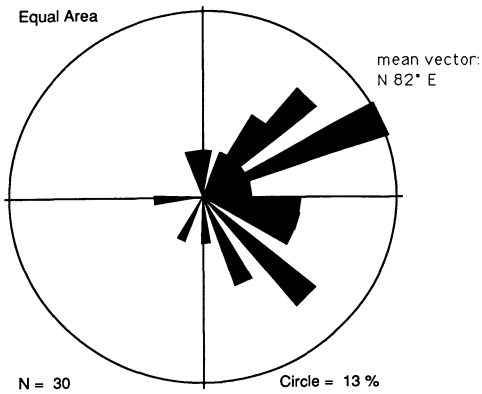


FIG. 6.—Rose diagram showing the paleoflow directions indicated by imbricated clasts at three stratigraphic levels in the El Corral Formation. Results of a larger data set, measured with no preference to operator perception of whether or not the clasts were imbricated, also show east-directed paleoflow.

Geometries and textures suggest that the El Corral Formation conglomerates were deposited from unconfined or little-confined flows, probably by sheetflood or in shallow, ephemeral braided streams (e.g., Nemec and Steel 1984; Wells 1984). These depositional processes are typical of proximal positions on an alluvial fan or bajada.

Volcanic-derived strata occur at several levels in the Río Azul section. In addition to three thin ash and bentonite beds sampled in the Cuculí Formation (fig. 3), bentonitic mudstones are common in the zone from about 900 to 1200 m. In the El Corral Formation, thick volcaniclastic sandstones, including dacitic boulders, are prominent (fig. 3). The lowest such bed (~2315 m) yielded one clean ash fall, from which euhedral zircons were hand picked for dating. The other beds, which form horizons as much as 20 m thick, are common at stratigraphic levels between

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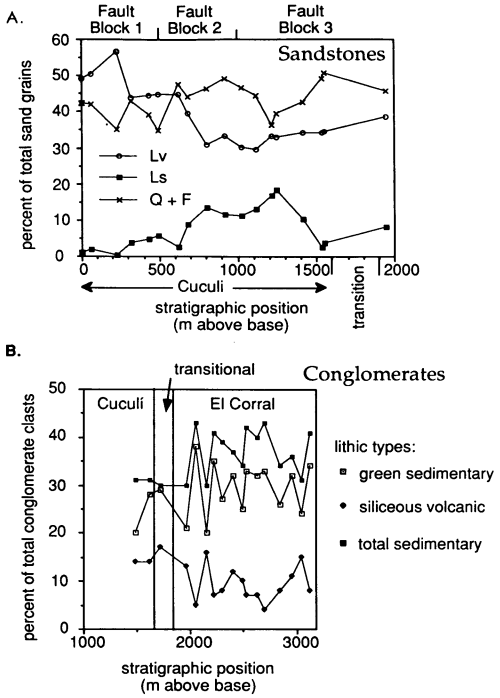


FIG. 7.—(*A*) Compositions of framework grains in sandstones from the Cuculí Formation, base of the El Corral Formation, and their transitional zone, based on petrographic determinations of at least 300 grains in a grid for each sample. Boundaries correspond to the fault blocks discussed in text. Lv-total volcanic lithic grains; Ls-total sedimentary lithic grains (excluding metasedimentary grains, which are not shown); Q + F-summed monocrystalline quartz and feldspar grains. (*B*) Compositions of gravel clasts in the uppermost Cuculí and throughout the El Corral formations. Values based on recording compositions of 50 randomly selected clasts at each level. See text for list of all compositional types. Sedimentary lithics, and especially green sandstone and siltstone, are typical of sources in the Precordillera thrust belt. Siliceous volcanic lithics are derived from the Frontal Cordillera (fig. 1).

2600 and 2800 m. In those thick units the volcanic material is all redeposited.

PROVENANCE

Our provenance determinations for the Cuculí Formation come largely from sandstones, whereas interpretation of El Corral Formation provenance is based largely on conglomerate compositions. Sixteen sandstones of the Cuculí Formation and one from the base of the El Corral Formation were examined with a petrographic microscope, and the compositions of a minimum of 300 grains per thin section were tabulated.

Sandstones throughout the Cuculí Formation are lithic arenites, with volcanic lithic grains dominating over sedimentary lithic grains (fig. 7*a*). Monocrystalline grains consist of quartz and feldspar; lithic grains include sedimentary rock fragments (shale, siltstone, and argillaceous mudstone), volcanic lithics (anhedral felsic grains with fine feldspar phenocrysts), and minor metasedimentary fragments (slates and phyllitic fragments). The overall abundance of lithic grains declines slightly through time while the abundance of monocrystalline grains increases (fig. 7*a*). Volcanic lithic clasts are most abundant in the lower 800 m of the section (ranging from about 40 to 55% of total framework grains), and then decline to a relatively stable value near 35% (fig. 7*a*). This trend is largely reciprocal with sedimentary lithics, whose maximum abundance is from about 800 to 1400 m in the section. No sand grains of limestone composition are found below 1000 m in the section, and even the sampled sandstones from higher in the section have <1% limestone clasts.

The conglomerate beds in the lower and middle Cuculí Formation have clasts rich in siliceous volcanics, quartz-rich metamorphics, green and pink sandstones, andesites, chert, and green fine-grained metasediments, with local intraclast-rich horizons. No limestone clasts were noted among the gravel-size fraction below 1550 m; higher levels in the Cuculí and El Corral formations have limestone clasts.

Gravel clast compositions in the El Corral Formation include green sedimentary lithics, siliceous volcanics, andesitic and dacitic volcanics, limestone, and red and white sandstone. Green sandstones dominate the El Corral (fig. 7*b*). The composition changes slightly from the zone transitional to the Cuculí Formation to the pure El Corral facies: the ratio of green sandstone clasts to siliceous volcanic clasts increases (fig. 7*b*).

The provenance of conglomerate clasts can be readily interpreted, and the sources of the sand grains can also be estimated from the regional geology (J. J. Zambrano unpub. map of San Juan province; Damanti 1989). The siliceous volcanic clasts of both formations are diagnostic of a source in the Permo-

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Triassic Choiyoi volcano/plutonic complex, which is exposed west of the Precordillera, in the Frontal Cordillera (fig. 1). Andesitic and dacitic volcanic clasts might be derived from either the Frontal Cordillera or from local centers in the Precordillera. Green low-grade metasedimentary clasts are also typical of the Frontal Cordillera or might be derived from the westernmost Precordillera. Green sandstones and siltstones are indicative of Silurian or Devonian source rocks, which are now exposed within the Precordillera, west of the Río Azul site. The Paleozoic sedimentary units are also likely sources of monocrystalline quartz and feldspar grains of the sand fraction. Limestone clasts are especially diagnostic of sources in the central part of the Precordillera; the limestone comes from Ordovician and Cambrian units exposed only by virtue of thrust faulting (e.g., Johnson et al. 1986). It is not certain whether the few sand-sized limestone grains present in the Cuculí Formation were derived from the Paleozoic limestones or from pedogenic carbonates; however it is clear that the limestone pebbles and cobbles were derived from the Paleozoic bedrock. The red and white sandstone clasts are typical of Carboniferous and Permian bedrock units of the Precordillera.

MAGNETIC POLARITY ZONATION AND FISSION TRACK AGES

Two types of magnetic behavior characteristic of Río Azul samples—single-component and multi-component magnetization—are revealed by alternating-field and thermal demagnetization studies. In the first, illustrated for site RA30, Zijderveld diagrams indicate that a secondary component was removed by heating to 200°C. At higher temperatures, decay toward the origin is established, indicating a normal polarity characteristic magnetization (fig. 8*c*). Intensity plots reveal coercivities of NRM dominantly below 400 Oe and blocking temperatures dominantly below 600°C, typical of remanence carried by magnetite. A second typical behavior is illustrated by site RA8, for which a small secondary component of intermediate orientation was removed by heating to 100°C or by an alternating-field demagnetization to 50 Oe. Progressive demagnetization then produced a linear decay trend up to 200 Oe or 400°C. As revealed by thermal demagnetization, this component had a northerly and upward direction and is likely a viscous component acquired parallel to the local magnetic field at the collecting locality. Above 200 Oe alternating-field demagnetization or above 400°C, a linear trend is established which defines a reversed polarity characteristic remanence orientation (fig. 8*b,d*). These samples display high coercivities and have a majority of blocking temperatures above 580°C (fig. 8*e,f*), suggesting that hematite carries the principal remanence.

Because thermal demagnetization proved successful for removing secondary magnetizations, a blanket thermal treatment of 400°C was used as the starting point for demagnetization of all samples. All but about 5% of the samples were heated to at least 500°C, and many of them to considerably higher temperatures to clarify the direction of the characteristic component (Rutty 1988) (e.g., 40% were heated to temperatures between 600° and 710°C).

Of the 114 sites whose magnetic polarity was measured, 89 yielded statistically significant data and are classified as Class I sites ($R \geq 2.63$ for $N = 3$; Watson 1956) (fig. 4*b*) and two produced uninterpretable results and are not included in the illustrations and discussions that follow. Of the 22 Class II sites (defined as $N < 3$ or $R < 2.63$), 12 have only two samples because one of the three samples was destroyed in transit or sample preparation, but the remaining two samples from those sites gave consistent results. Of the other 10 Class II sites, five are located in the lower 110 m of the section. Two factors might contribute to more complex magnetization at the base of the section: pedogenesis and difficulty preparing the siliceous, highly indurated samples typical of that part of the section. As noted in the lithologic description, moderate to severe soil alteration was common near the base of the Cuculí Formation, potentially leading to complex remagnetization of samples collected in paleosols.

The site-mean directions for Class I sites are shown in figure 9; here we exclude sites with Virtual Geomagnetic Pole (VGP) latitudes transitional between normal and reversed polarity zones (fig. 4*b*). Because the means of normal polarity sites and the reversed polarity sites are nearly antipodal, we conclude that the reversals test is passed and

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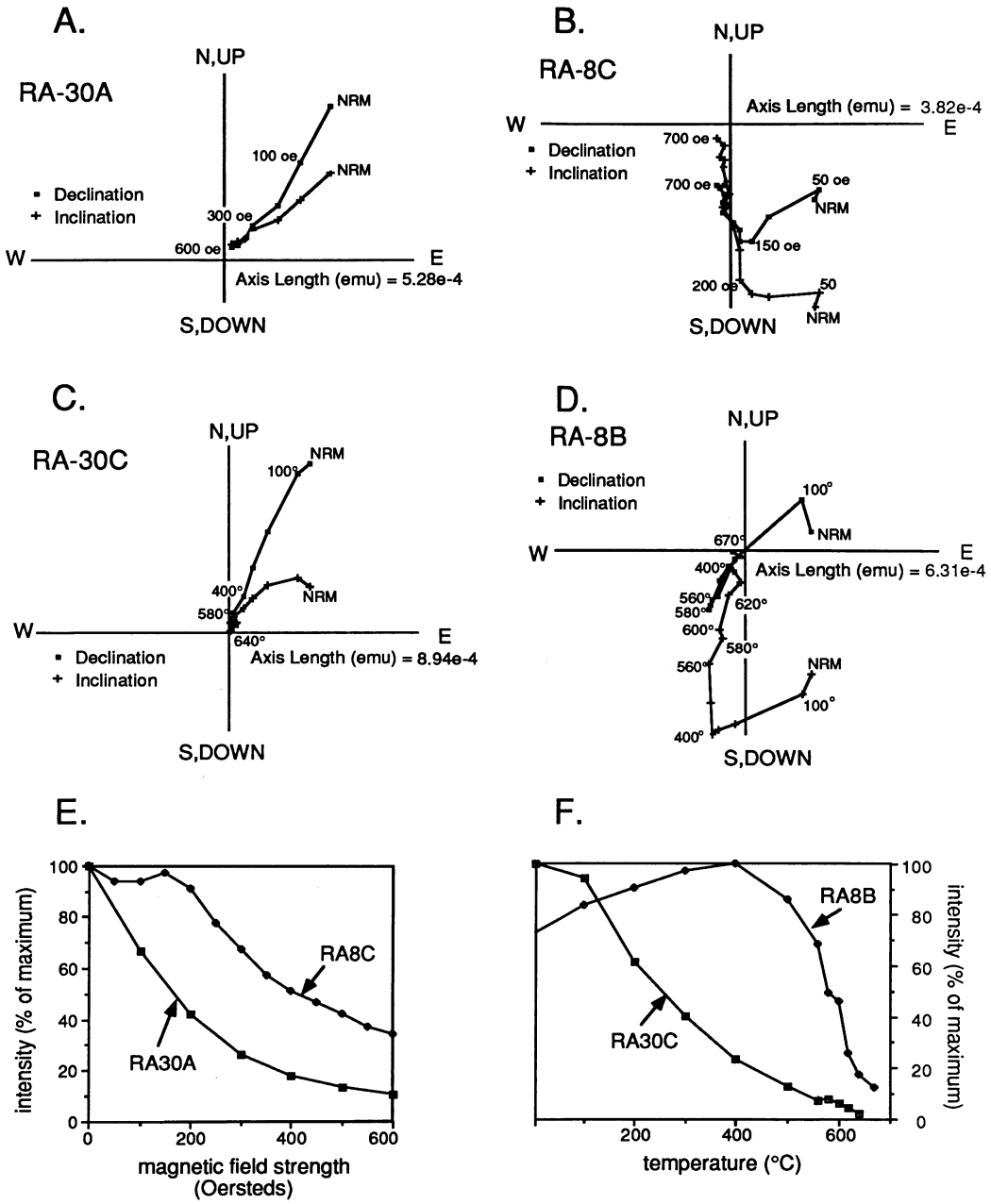


FIG. 8.—Graphs showing behavior of representative samples from Río Azul during stepwise progressive demagnetization, *A*, *B*, *C*, and *D* are Zijderveld diagrams, which simultaneously illustrate paleomagnetic intensity (as distance from the origin of the plot) and vector orientation, at successive demagnetization steps. *A* and *B* illustrate alternating-field demagnetization: *A* in 100 Oe steps from 0 to 600 Oe, and *B* in 50 Oe steps from 0 to 700 Oe. *C* and *D* show thermal demagnetization: *C* in 100°C steps from 0 to 500°C, then in 20°C steps from 560° to 640°C; *D* in steps equal to those of *C* with the addition of 670°C measurement. *E* and *F* show intensity of magnetization at progressive demagnetization steps; *E* for alternating field demagnetization and *F* for thermal demagnetization. Site RA30 (*A*, *C*, and indicated lines in *E* and *F*) illustrates behavior typical of samples with a single component of magnetization, probably carried by magnetite. Site RA8 (*B*, *D*, and indicated lines in *E* and *F*) illustrates behavior typical of samples with multiple components of magnetization, probably carried by hematite.

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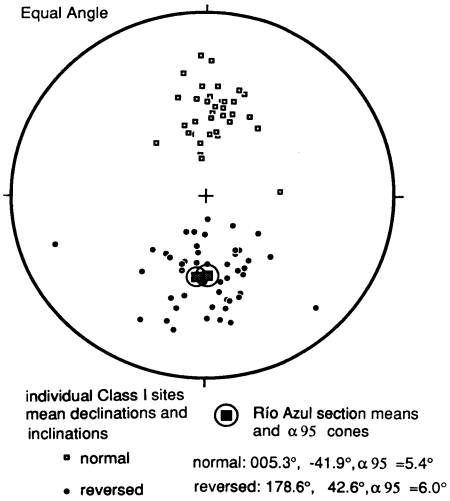


FIG. 9.—Equal-angle projection of site-mean directions of normal sites (open symbols) and reversed sites (closed symbols), excluding Class II sites and three sites with low VGP latitudes located at transitions between polarity zones. Open boxes are upper hemisphere projections, filled circles are lower hemisphere projections. The means of all normal and all reversed sites are shown by 95% confidence cones; the mean of the normal sites was inverted to the lower hemisphere to show that it is nearly antipodal to the mean of the reversed-polarity sites.

therefore that the data (fig. 4*b*) reveal a stable ancient magnetization.

Twenty-nine magnetic polarity zones are defined (fig. 4*b*), of which one near the top of the section and two at the base of fault block 2 are single-site polarity zones defined by Class I sites. A greater thickness of the section has reversed polarity than normal polarity (fig. 4*c*). A class I site at 1739 m, which yields a low VGP latitude, has not been interpreted to constitute a polarity zone (fig. 4*b*). Its demagnetization data are too irregular to warrant erecting a single-site polarity zone for that site, and in turn creating another single-site polarity zone immediately beneath it.

Because of the three fault zones that break the Río Azul section, we cannot make the assumption that the composite pattern of magnetic zones should correlate to the magnetic polarity time scale. The faults might either repeat or omit section and polarity zones. We evaluate, therefore, the correlation of the magnetic zones to the magnetic polarity time scale independently for each fault block. The mean directions of all reversed and all normal polarity sites within each fault block were plotted on equal-angle projections (not illustrated). No significant difference in the mean directions from the various fault blocks was observed, implying no relative rotation during local deformation.

Three air-fall volcanic ashes in the section yielded sufficient quantities of euhedral zircon crystals to be dated by the external detector fission track method (table 1). Two of the dated ashes occur in the part of the section for which magnetic polarity zonation was established; the third sample is from the conglomeratic part of the section in which no materials suitable for paleomagnetic measurement occur (figs. 3, 4). The ages range from 12.5 ± 2.0 Ma to 9.2 ± 2.5 Ma, indicating that the Río Azul section is probably entirely middle and late Miocene in age. Because of relatively low uranium contents in the zircons (table 1), the 2-σ errors on the ages are on the order of 2 to 2.5 m.y. Inadequate quantities of biotite were extracted from the samples to permit any attempt to date them by more precise radiometric methods.

TABLE 1

FISSION TRACK DATA FOR ZIRCONS SEPARATED FROM VOLCANIC ASHES AND BENTONITES INTERCALATED IN RÍO AZUL STRATA

Sample	Total grains	Fossil track density (10^5 tks/cm^2)	Induced track density (10^5 tks/cm^2)	Age (Ma)	Error (2 σ)
RA1	12	13	59	12.5	2.0
RA3	9	11	52	10.4	2.1
RA4	8	4.6	25	9.2	2.5

NOTE.—Constants: total decay constant for ^{238}U = 1.55e-10/yr; decay constant for spontaneous fission of ^{238}U = 7.03e-17/yr; cross section for thermally induced fission of ^{235}U = 580e24cm^2; isotopic ratio ^{235}U/^{238}U = 7.252e-3.

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CHRONOSTRATIGRAPHIC CORRELATION OF THE RIO AZUL SECTION

Fission track ages on intercalated ash beds demonstrate that the Río Azul section is of middle and late Miocene age (table 1), but there is more than one possible correlation of the local magnetic polarity pattern to the magnetic polarity time scale. This situation is principally due to faults that break the section into thin segments, some of which have few polarity zones and little independent information on which to base a correlation. The difficulty is compounded by uncertainties of approximately 2 m.y. on the ages of intercalated ashes, the existence of paleosols, and the fact that the middle and early-late Miocene magnetic polarity time scale is characterized by numerous short polarity intervals.

Information in addition to the pattern of reversals allows us to eliminate many possible correlations and to interpret the most likely correlations. Correlations of the magnetic polarity zones of the lower (1) and upper (3) fault blocks are constrained by the fission track dates on intercalated ashes. A strong additional limitation comes from the fact that the local magnetic polarity pattern (fig. 4*c*) contains no outstandingly long normal polarity zone likely to correlate to the long normal period in chron C5 (8.9 to 10.4 Ma) (Berggren et al. 1985). No independent date is available for the strata of the middle fault block (2). We noted no consistent lithologic patterns in the Cuculí Formation that would allow us to independently correlate strata across the faults. The lithologic similarity suggests that strata of similar ages may be present in all the fault blocks and that the offset on the faults was not more than a few tens or hundreds of meters.

Rocks of fault block 1 are dominantly of reversed polarity, and the uppermost reversed zone contains an ash whose age is 12.5 ± 2.0 Ma (fig. 10*a*). Given the error on the fission track date and the fact that it occurs in rocks of reversed polarity, the ash can be no older than the base of chron C5AC (14.2 Ma). Whereas the polarity zonation of the upper 350 m of fault block 1 is probably complete, the four lower polarity zones, which occur in strata with six major pedogenic horizons, may well be very incomplete records of many polarity zones representing several million years. Based on the radiometric date and on the fact that the upper 350 m of the lower fault block lack soils which might indicate long hiatuses in deposition, we correlate the upper part to the interval spanning the lower part of chron C5A (13 to 12.2 Ma). With this correlation, the base of the section can be no younger than the base of chron C5AB (13.7 Ma), but we speculate that the base of the section could be as old as 14 to 17 Ma (based on comparison to other dated sections in the region (Reynolds et al. 1990; Tabbutt et al. 1989).

"Fault block 2" is in fact internally discontinuous, with a pair of faults between 865 and 890 m. However, there is too little chronologic information available to warrant further subdivision. Below 865 m the eight polarity zones include a greater thickness of strata of reversed polarity than of normal polarity (fig. 10*b*). Based largely on the relative thicknesses of the zones and the slight dominance of reversed polarity, we suggest a correlation to parts of chrons C5A through C5AB (about 12.5 and 13.5 Ma). The greatest apparent difficulty with this correlation comes from the discrepancy between a long normal interval in chron C5AC (13.7–14.1 Ma) in the Berggren et al. (1985) time scale and a pair of very short polarity zones in the local paleomagnetic column (the lowest normal and reversed zones in fault block 2). However, those two polarity zones are defined by single Class I sites collected within 1 m of the fault zone. The sites might represent a brief fluctuation in the Miocene earth's field, structural disruption and mixing along the fault, or errors in our identifications of the relative positions of the two sites.

The correlation of the upper part of fault block 2 is even more uncertain, given the small number of reversals (figs. 4*b*, 10*b*). If there was little offset on the fault at 865 m, and if the rates of accumulation are comparable to those of the lower part of fault block 2, the thicknesses of normal and reversed zones suggest a correlation to the upper part of chron C5A (interval from 11.7 to 12.1 Ma). The correlations of both parts of fault block 2 are of course non-unique, given the available information.

The correlation of the 1200 m thick part of fault block 3 for which we have paleomagnetic data (figs. 4*b*, 10*c*) is constrained by an

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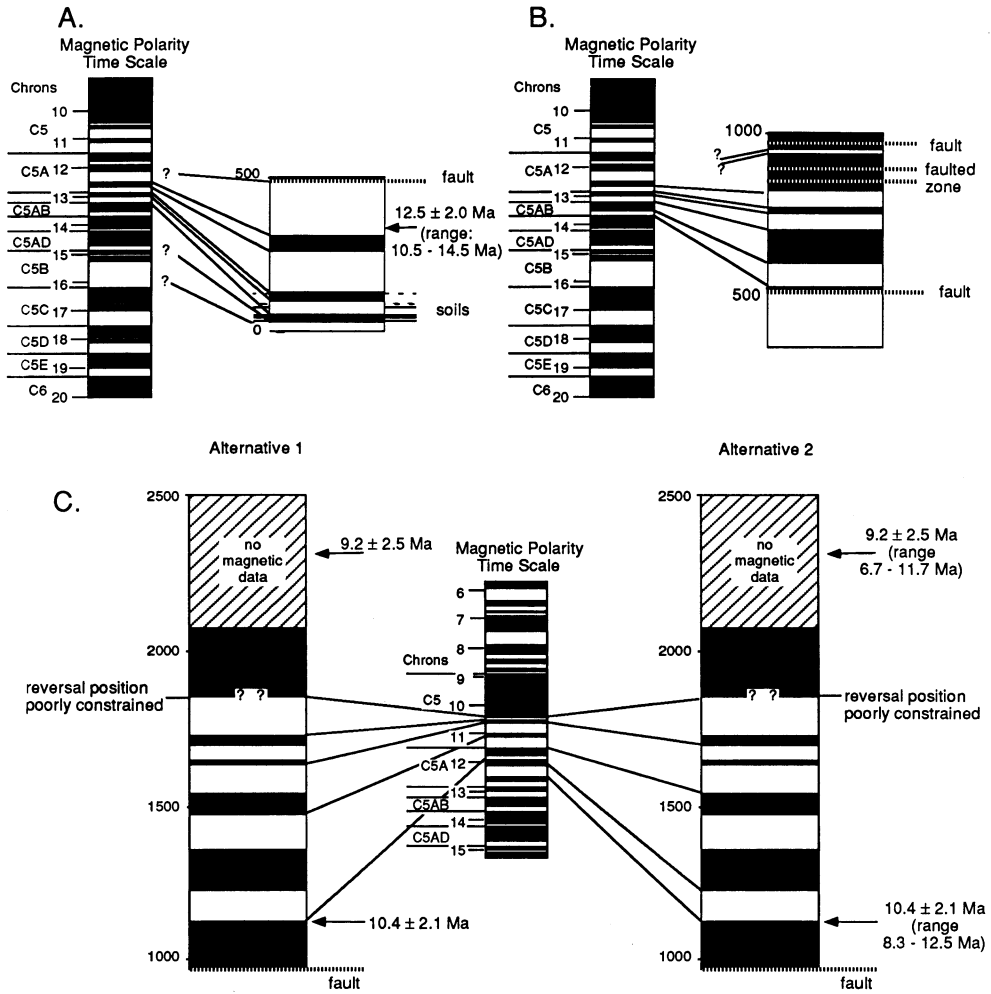


FIG. 10.—Correlations of Río Azul magnetic polarity zones to the magnetic polarity time scale (Berggren et al. 1985). Correlation of each of the three main fault blocks is treated independently. (*A*) fault block 1 (lower part of section); (*B*) fault block 2 (middle part of section); and (*C*) two alternative correlations of fault block 3, including overlying strata for which there are not magnetic polarity zonation data.

ash (10.4 ± 2.1 Ma) at the top of the lowest normal polarity zone and by an ash (9.2 ± 2.5 Ma) in the conglomerates above the zone with paleomagnetic data. Thus the strata between the ash beds should be no older than 12.5 Ma and no younger than 6.7 Ma. The upper part of the section with magnetic data is largely of reversed polarity, and the lower part has sub-equal amounts of normal and reversed polarity. What appears to be the uppermost thick normal zone (fig. 10*c*) is constrained by a single normal site that lies 220 m above the next highest site (fig. 4*b*) (there are no intervening sites because of coarse lithologies). With the possible exception of this uppermost normal zone, we suggest that none of the other normal zones are sufficiently thick (compared to thicknesses of polarity zones throughout the Río Azul section) to represent chron C5 of the magnetic polarity time scale between 8.9 and 10.4 Ma (fig. 10*c*) (Berggren et al. 1985). This constraint, the fission track ages, and the pattern of the polarity zones suggest a correlation between ~10 and ~12.5 Ma for the part of fault block 3 with magnetic data.

However, at least two alternative interpretations of the details of this correlation are

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possible (fig. 10*c*). One interpretation (left side fig. 10*c*) correlates the fault-block 3 section to the lower part of chron C5 and upper part of chron C5A. This correlation suggests that the Río Azul section captures a brief reversed polarity zone not shown in the magnetic polarity time scale (embedded within the normal interval that spans 10.54 to 10.59 Ma) (Berggren et al. 1985). The existence of such brief polarity zones is not unusual in regions of rapid sediment accumulation (e.g., Beer 1990), but the data presented here are not sufficient to insure that the single-site polarity zone in question is truly such an added detail rather than an error. An alternative correlation (fig. 10*c*, right) interprets the single-site normal zone as a major polarity zone and shifts the correlation further back in time, such that the section spans the lower part of chron C5 through much of chron C5A. Using the Berggren et al. (1985) time scale, this correlation places the dated ash very close to 12.5 Ma, which is the lower limit of the zircon fission track determination. The major difficulty with either of these correlations is that the lowest normal interval in the local polarity column is relatively thick, unlike the short interval of normal polarity to which we have correlated it (fig. 10*c*, right). If either correlation is accurate, it implies an anomalously high rate of accumulation during the basal normal interval. We note this interval coincides with unusually bentonitic mudstones in the section and speculate that volcanic activity might have caused a higher-than-average rate of accumulation.

The age of the upper 1000 m of fault block 3 (fig. 4) cannot be determined directly. Neither lithologic discontinuities nor geometrically defined unconformities occur. Hence we interpret sediment accumulation to have been continuous with that in the lower part of the upper fault block. The age of the top of the section cannot be predicted without independent knowledge of the rates of accumulation. Nevertheless, given the dated ash, the lack of evidence of unconformities, and a prejudice that the high-energy basin-margin conglomerate facies might suggest high rates of accumulation; we speculate that the top of the Río Azul section is roughly 8 to 9 Ma.

Compared to correlations possible elsewhere in the Bermejo basin (e.g., Johnson et al. 1986; Beer 1990; Reynolds et al. 1990; Milana, Bercowski and Jordan unpub. data), the details of the correlations of the Río Azul magnetic polarity zonation to the magnetic polarity time scale are uncertain. Given this lack of detail, construction of a stratal accumulation curve does not seem warranted. The proposed alternative correlations of fault block 3 (fig. 10*c*) indicate an average accumulation rate of ~0.4 to 0.6 mm/yr; that for the upper part of fault block 1 (fig. 10*a*) accumulated at a mean rate of ~0.4 mm/yr. These rates are within the range of values confirmed elsewhere in the Bermejo basin.

SUMMARY, PALEOGEOGRAPHIC RECONSTRUCTION, AND DEFORMATION HISTORY

The faults in the Río Azul section preclude the assumption of continuity of accumulation that is inherent in attempts to correlate a local magnetic polarity stratigraphy to the magnetic polarity time scale. Nevertheless, by combining polarity patterns of individual fault blocks with independent constraints on their ages we have been able to make general correlations of the section. Because the available chronologic data exclude many possible correlations and because the polarity patterns resemble those of certain parts of the magnetic polarity time scale, we conclude that our general correlations are reliable.

The correlations indicate that the base of the Cuculí Formation is as old as 15 to 17 Ma (and is certainly older than 13.6 Ma). The ~300 m thick transitional zone between Cuculí and El Corral Formations spans roughly 10.8 (or 10.6, for alternative 1)–10.4 Ma. The faulted top of the El Corral Formation is probably no younger than 8 to 9 Ma. Correlation to the Huaco section (figs. 1, 11) illustrates that the Miocene Bermejo basin thickened westward. The proposed correlations imply that 500 m of the Río Azul section are repeated across the lower fault, and that a smaller duplication of section, roughly 100 to 200 m, occurs across the combination of faults within and bounding the top of the second fault block (fig. 11).

The bed geometries in the Cuculí Formation are similar to those of the Jarillal Formation exposed at Huaco and suggest deposition on an alluvial plain with ephemeral low-order streams. At Río Azul the Cuculí spans from about 15–17 Ma to ~10.8 (or 10.6) Ma; at Huaco the lower part of the Jarillal Forma-

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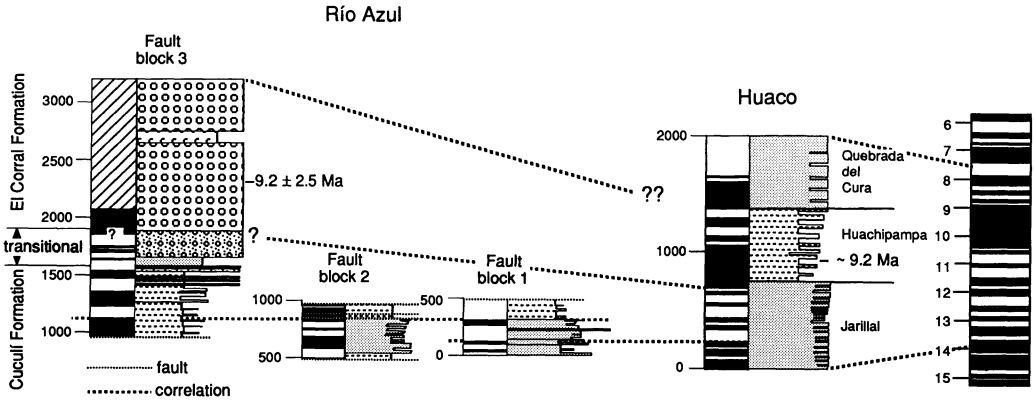


FIG. 11.—Comparison of magnetic polarity zonation and lithostratigraphy of the Río Azul fault-bounded sections and the lower part of the Huaco section (see fig. 1 for locations and fig. 4 for lithologic symbols), utilizing correlation alternative 2 of the upper fault block at Río Azul (fig. 10*c*, right side). The correlation of the three Río Azul sections is hung on the upper boundary of a common normal zone; that normal zone has not been measured in the Huaco section (Johnson et al. 1986); hence the correlation to Huaco is hung on the top of the next-older normal zone.

tion is covered, but the exposed part spans from 14 to ~10 Ma (Johnson et al. 1986; Beer 1990). Much of the Cuculí Formation is thus correlative with the Jarillal Formation (fig. 11). Based on paleoflow directions at Huaco and preliminary paleocurrent data for Río Azul, Río Azul strata accumulated upslope of those at Huaco (fig. 12*a*).

Sheetflood sandstones and siltstones of the Huachipampa Formation of Huaco formed contemporaneously with the sheet geometry coarse conglomerates of the El Corral Formation of Río Azul, and both are interpreted to represent deposition from unconfined flows on an alluvial fan or bajada. The Huachipampa Formation spans 10–8.4 Ma; we estimate the El Corral to span 10.4–8 or 9 Ma, with the top faulted. At both locations paleoflow was to the east, implying that El Corral could be the proximal facies of the alluvial fan on which Huachipampa facies were distal deposits (Beer and Jordan 1989). Contrasting clast compositions of the two correlative units suggest, however, that they did not form on a single fan: Precordillera-sourced clasts dominate the proximal El Corral (fig. 7*b*), whereas the Huachipampa has a strong influence of Frontal Cordillera clasts in addition to Precordillera material (Damanti 1989).

At Huaco, Jarillal-type deposits are replaced by the Huachipampa Formation across a transition that spanned about 400,000 yr and was complete by 10 Ma (Beer and Jordan 1989; Beer 1990). The facies change represents a change from an alluvial plain consisting of many small, sinuous, east-flowing streams during Jarillal time (fig. 12*a*) to sheetflood deposits of a single major alluvial fan during Huachipampa time (fig. 12*b*). Based upon clast compositions and facies characteristics, this paleogeographic change is attributed to the growth of a topographic high in the Precordillera west of Huaco, which significantly changed regional drainage patterns (Damanti et al. 1988; Beer and Jordan 1989). Likewise at Río Azul, important changes in the drainage system occurred between the low-order streams of the Cuculí Formation and the proximal fan(s) of the El Corral Formation. A distal source of rivers and sediment was replaced by a much more proximal source area; the change of sand-clast and gravel-clast composition indicates that sources in the distant Frontal Cordillera were cut off, leaving only sources in the Precordillera west of Río Azul. We interpret that the Cuculí Formation represents more proximal deposits of the same alluvial plain as the Jarillal Formation, but that the El Corral deposits formed a bajada fringing the rising Precordillera and were not part of the large point-source fan that generated the Huachipampa Formation (fig. 12*b*). The facies transition zone at Río Azul, whose age is estimated to span ~10.8 (or 10.6) to 10.4 Ma, corresponds well in character to the

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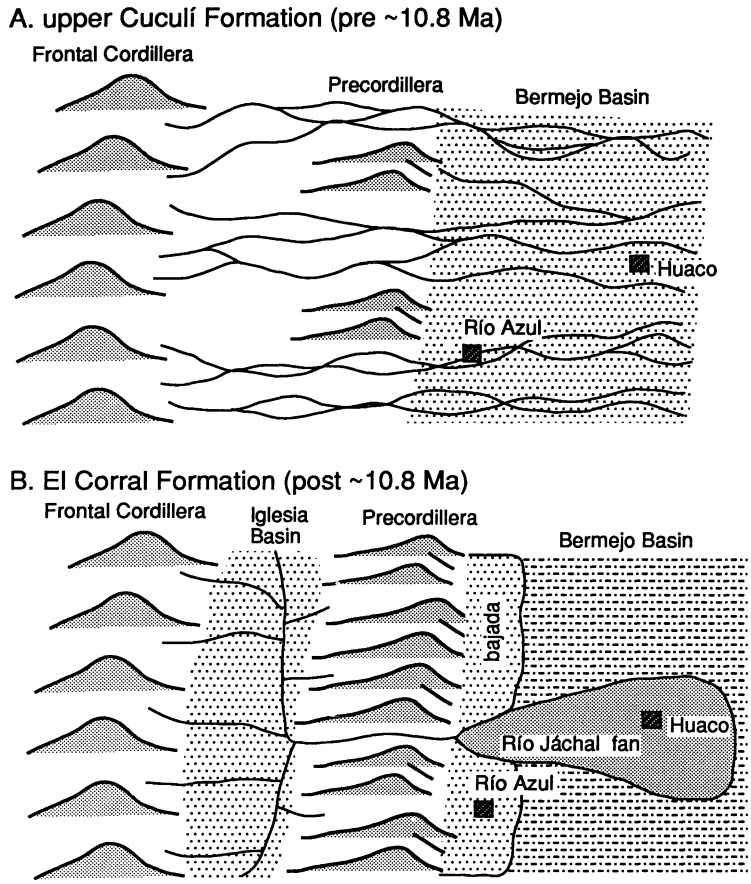


FIG. 12.—Paleogeographic evolution of the western margin of the Bermejo basin from ~13 to 9 Ma. (*A*) At the time of deposition of the upper Cuculí Formation, both Río Azul and Huaco were receiving sediment from the Frontal Cordillera in low-order streams, although local sources in the Precordillera were being eroded. (*B*) El Corral Formation deposits are of proximal character and show a dominance of Precordillera sources, whereas Huaco strata indicate deposition in a distal part of a large alluvial fan whose headwaters carry a mixture of sediment from the Frontal Cordillera and Precordillera (Damanti et al. 1988; Beer and Jordan 1989; Damanti 1989).

paleogeographic change envisioned for the Huaco strata and correlates approximately with the Jarillal-Huachipampa transition at Huaco.

It is clear from comparison of facies and clast compositions that there was a lag time between first thrusting in the central Precordillera and the major reorganization of the drainage system that caused the Cuculí/El Corral and Jarillal/Huachipampa boundaries. Sandstones older than ~13 Ma are rich in volcanic lithic grains, whereas younger sandstones have higher percentages of sedimentary lithics. This change reflects a shift from provenance dominated by the Frontal Cordillera to a Precordillera souce, with likely recycling of Frontal Cordillera clasts. The earliest limestone clast recognized to date that certainly reflects thrusting in the Precordillera, in the gravel-size fraction, occurs at ~11.3 (or 10.8) Ma. These compositional changes, demonstrating the shift of source area into the Precordillera, suggest that deformation of the Precordillera began at least 2 m.y. earlier than was expressed in the facies.

Knowledge of the ages of deformation in the Precordillera has been improved by study of the Río Azul thrust sliver in the proximal part of the Bermejo foreland basin. Crosscutting relations indicate that final motion on the thrust fault at the western edge of the Río Azul section (fig. 2, fault B) is younger than

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about 8 to 9 Ma. Deformation that rotated the entire Río Azul section and the underlying units, including the thrust fault that bounds the eastern edge of the structural sliver (fig. 2, fault A), about a horizontal axis is entirely younger than 8–9 Ma. Indirect constraints on ages of deformation come from the paleogeographic interpretations. Thrusting that led to uplift of Paleozoic strata in the drainage basin to the west of the Río Azul sliver began at this latitude no later than ~13 Ma and had created a sufficiently continuous ridge line by 10.8 (or 10.6) Ma that the drainage system changed drastically. The direct and indirect age constraints indicate a west-to-east progression of the eastern edge of Precordillera deformation, but do not rule out continued (i.e., out-of-sequence) deformation to the west of the eastern front. Although the age of earliest deformation proves to be older than was suspected based only on Huaco strata (Johnson et al. 1986; Beer and Jordan 1989), it is still apparently younger than the 16 Ma indicated 150 km further north at Las Juntas (fig. 1) (Reynolds et al. 1990; Damanti 1989).

Viewed in isolation, the history of the Río Azul section and thrust faults to which it is related is a useful addition to knowledge of one element of the growth of the Andes Mountains. The Río Azul results provide us with considerable insight into controls on the stratigraphy of more distal sites in the Bermejo basin, such as at Huaco and Mogna, and thus are of great value in our long-term study of the dynamic history of the Bermejo foreland basin. Furthermore, our clarification of the degree to which dramatic facies changes lag behind the initiation of thrusting in a mid-to-proximal location in a foreland basin can probably be generalized to other nonmarine foreland basins.

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